

Memory Coalescing

- Threads in the same warp should pull from consecutive memory addresses
- Sort of like cache locality

Global Memory

- Every warp in block has to pull from same memory
 - Warp 0 →
 - ⋮ → pulls same global memory
 - Warp n →
- Keep one line of data here *inefficient*

Shared mem

- Block loads once into shared memory *shared per block!*
 - Warp 0 →
 - ⋮ → pulls from shared (faster)
 - Warp n →
- Keep reused data here

Hierarchy - fast to slow

← try to optimize for this

- Registers → single values used by 1 thread
- Shared mem → data reused in a block
- L1/L2 cache → automatic, local data
- Global memory → data only used once

Streaming Multiprocessor (SM)

- Contains
 - Cuda Cores (ALUs)
 - Warp Schedulers
 - Registers
 - Tensor Cores
- Shared memory
- Caches
- Assigned blocks to work on